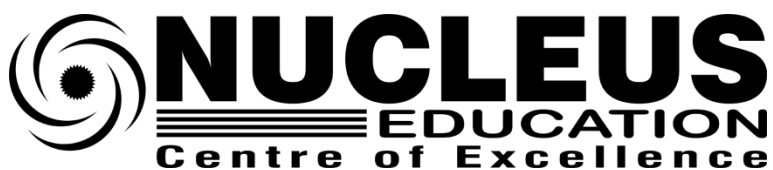


IIT MATHEMATICS

BINOMIAL THEOREM



BINOMIAL THEOREM

1. BINOMIAL EXPRESSION :

Any algebraic expression which contains two dissimilar terms is called binomial expression.

For example : $x - y$, $xy + \frac{1}{x}$, $\frac{1}{z} - 1$, $\frac{1}{(x-y)^{1/3}} + 3$ etc.

2. BINOMIAL THEOREM :

The formula by which any positive integral power of a binomial expression can be expanded in the form of a series is known as **BINOMIAL THEOREM**.

If $x, y \in \mathbb{R}$ and $n \in \mathbb{N}$, then :

$$(x + y)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} y + {}^nC_2 x^{n-2} y^2 + \dots + {}^nC_r x^{n-r} y^r + \dots + {}^nC_n y^n = \sum_{r=0}^n {}^nC_r x^{n-r} y^r$$

This theorem can be proved by induction.

Observations :

- (a) The number of terms in the expansion is $(n + 1)$ i.e. one more than the index.
- (b) The sum of the indices of x & y in each term is n .
- (c) The binomial coefficients of the terms (${}^nC_0, {}^nC_1, \dots$) equidistant from the beginning and the end are equal. i.e. ${}^nC_r = {}^nC_{n-r}$
- (d) Symbol nC_r can also be denoted by $\binom{n}{r}$, $C(n, r)$ or A_r^n .

Some important expansions :

- (i) $(1 + x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$.
- (ii) $(1 - x)^n = {}^nC_0 - {}^nC_1 x + {}^nC_2 x^2 + \dots + (-1)^n \cdot {}^nC_n x^n$.

Note : The coefficient of x^r in $(1 + x)^n = {}^nC_r$ & that in $(1 - x)^n = (-1)^r \cdot {}^nC_r$

Illustration 1 : Expand : $(y + 2)^6$.

Solution : ${}^6C_0 y^6 + {}^6C_1 y^5 \cdot 2 + {}^6C_2 y^4 \cdot 2^2 + {}^6C_3 y^3 \cdot 2^3 + {}^6C_4 y^2 \cdot 2^4 + {}^6C_5 y^1 \cdot 2^5 + {}^6C_6 \cdot 2^6$
 $= y^6 + 12y^5 + 60y^4 + 160y^3 + 240y^2 + 192y + 64.$

Illustration 2 : Write first 4 terms of $\left(1 - \frac{2y^2}{5}\right)^7$

Solution : ${}^7C_0, {}^7C_1 \left(-\frac{2y^2}{5}\right), {}^7C_2 \left(-\frac{2y^2}{5}\right)^2, {}^7C_3 \left(-\frac{2y^2}{5}\right)^3$

Illustration 3 : If in the expansion of $(1 + x)^m (1 - x)^n$, the coefficients of x and x^2 are 3 and -6 respectively then m is - **[JEE 99]**

- (A) 6 (B) 9 (C) 12 (D) 24

Solution : $(1+x)^m (1-x)^n = \left[1+mx + \frac{(m)(m-1).x^2}{2} + \dots \right] \left[1-nx + \frac{n(n-1)}{2} x^2 + \dots \right]$

Coefficient of $x = m - n = 3$ (i)

Coefficient of $x^2 = -mn + \frac{n(n+1)}{2} + \frac{m(m-1)}{2} = -6$ (ii)

Solving (i) and (ii), we get
 $m = 12$ and $n = 9$.

Do yourself - 1 :

- (i) Expand $\left(3x^2 - \frac{x}{2}\right)^5$ (ii) Expand $(y+x)^n$

Pascal's triangle :

$(x+y)^0$	1	1
$(x+y)^1$	x + y	1 1
$(x+y)^2$	$x^2 + 2xy + y^2$	1 2 1
$(x+y)^3$	$x^3 + 3x^2y + 3xy^2 + y^3$	1 3 3 1
$(x+y)^4$	$x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^5$	1 4 6 4 1

Pascal's triangle

- Pascal's triangle** - A triangular arrangement of numbers as shown. The numbers give the binomial coefficients for the expansion of $(x+y)^n$. The first row is for $n = 0$, the second for $n = 1$, etc. Each row has 1 as its first and last number. Other numbers are generated by adding the two numbers immediately to the left and right in the row above.
- Pascal triangle is formed by binomial coefficient.
- The number of terms in the expansion of $(x+y)^n$ is $(n+1)$ i.e. one more than the index.
- The sum of the indices of x & y in each term is n .
- Power of first variable (x) decreases while of second variable (y) increases.
- Binomial coefficients are also called **combinatorial coefficients**.
- Binomial coefficients of the terms equidistant from the beginning and end are equal.
- r^{th} term from the beginning in the expansion of $(x+y)^n$ is same as r^{th} term from end in the expansion of $(y+x)^n$.
- r^{th} term from the end in $(x+y)^n$ is **$(n-r+2)^{\text{th}}$ term from the beginning**.

3. IMPORTANT TERMS IN THE BINOMIAL EXPANSION :

- (a) **General term:** The general term or the $(r+1)^{\text{th}}$ term in the expansion of $(x+y)^n$ is given by $T_{r+1} = {}^nC_r x^{n-r} y^r$.

Illustration 4 : Find : (a) The coefficient of x^7 in the expansion of $\left(ax^2 + \frac{1}{bx}\right)^{11}$
(b) The coefficient of x^{-7} in the expansion of $\left(ax - \frac{1}{bx^2}\right)^{11}$

Also, find the relation between a and b , so that these coefficients are equal.

Solution :(a) In the expansion of $\left(ax^2 + \frac{1}{bx}\right)^{11}$, the general term is :

$$T_{r+1} = {}^{11}C_r (ax^2)^{11-r} \left(\frac{1}{bx}\right)^r = {}^{11}C_r \cdot \frac{a^{11-r}}{b^r} \cdot x^{22-3r}$$

putting $22 - 3r = 7$

$$\therefore 3r = 15 \Rightarrow r = 5$$

$$\therefore t_6 = {}^{11}C_5 \frac{a^6}{b^5} \cdot x^7$$

Hence the coefficient of x^7 in $\left(ax^2 + \frac{1}{bx}\right)^{11}$ is ${}^{11}C_5 a^6 b^{-5}$.

Ans.

Note that binomial coefficient of sixth term is ${}^{11}C_5$.

(b) In the expansion of $\left(ax - \frac{1}{bx^2}\right)^{11}$, general term is :

$$T_{r+1} = {}^{11}C_r (ax)^{11-r} \left(\frac{-1}{bx^2}\right)^r = (-1)^r {}^{11}C_r \frac{a^{11-r}}{b^r} \cdot x^{11-3r}$$

putting $11 - 3r = -7$

$$\therefore 3r = 18 \Rightarrow r = 6$$

$$\therefore T_7 = (-1)^6 \cdot {}^{11}C_6 \frac{a^5}{b^6} \cdot x^{-7}$$

Hence the coefficient of x^{-7} in $\left(ax - \frac{1}{bx^2}\right)^{11}$ is ${}^{11}C_6 a^5 b^{-6}$.

Ans.

Also given :

$$\text{Coefficient of } x^7 \text{ in } \left(ax^2 + \frac{1}{bx}\right)^{11} = \text{coefficient of } x^{-7} \text{ in } \left(ax - \frac{1}{bx^2}\right)^{11}$$

$$\Rightarrow {}^{11}C_5 a^6 b^{-5} = {}^{11}C_6 a^5 b^{-6}$$

$$\Rightarrow ab = 1 \quad (\because {}^{11}C_5 = {}^{11}C_6)$$

which is the required relation between a and b.

Ans.

Illustration 5 : Find the number of rational terms in the expansion of $(9^{1/4} + 8^{1/6})^{1000}$.

Solution : The general term in the expansion of $(9^{1/4} + 8^{1/6})^{1000}$ is

$$T_{r+1} = {}^{1000}C_r \left(9^{1/4}\right)^{1000-r} \left(8^{1/6}\right)^r = {}^{1000}C_r 3^{\frac{1000-r}{2}} 2^{\frac{r}{2}}$$

The above term will be rational if exponents of 3 and 2 are integers

It means $\frac{1000-r}{2}$ and $\frac{r}{2}$ must be integers

The possible set of values of r is {0, 2, 4,, 1000}

Hence, number of rational terms is 501

Ans.

(b) Middle term :

The middle term(s) in the expansion of $(x + y)^n$ is (are) :

- (i) If n is even, there is only one middle term which is given by $T_{(n+2)/2} = {}^nC_{n/2} \cdot x^{n/2} \cdot y^{n/2}$
 (ii) If n is odd, there are two middle terms which are $T_{(n+1)/2}$ & $T_{[(n+1)/2]+1}$

Important Note :

Middle term has greatest binomial coefficient and if there are 2 middle terms their coefficients will be equal.

$$\Rightarrow {}^nC_r \text{ will be maximum } \begin{cases} \text{When } r = \frac{n}{2} \text{ if } n \text{ is even} \\ \text{When } r = \frac{n-1}{2} \text{ or } \frac{n+1}{2} \text{ if } n \text{ is odd} \end{cases}$$

$\Rightarrow$ The term containing greatest binomial coefficient will be middle term in the expansion of $(1 + x)^n$

Illustration 6 : Find the middle term in the expansion of $\left(3x - \frac{x^3}{6}\right)^9$

Solution : The number of terms in the expansion of $\left(3x - \frac{x^3}{6}\right)^9$ is 10 (even). So there are two middle terms.

i. e. $\left(\frac{9+1}{2}\right)^{\text{th}}$ and $\left(\frac{9+3}{2}\right)^{\text{th}}$ are two middle terms. They are given by T_5 and T_6

$$\therefore T_5 = T_{4+1} = {}^9C_4 (3x)^5 \left(-\frac{x^3}{6}\right)^4 = {}^9C_4 3^5 x^5 \cdot \frac{x^{12}}{6^4} = \frac{9 \cdot 8 \cdot 7 \cdot 6}{1 \cdot 2 \cdot 3 \cdot 4} \cdot \frac{3^5}{2^4 \cdot 3^4} x^{17} = \frac{189}{8} x^{17}$$

$$\text{and } T_6 = T_{5+1} = {}^9C_5 (3x)^4 \left(-\frac{x^3}{6}\right)^5 = -{}^9C_4 3^4 x^4 \cdot \frac{x^{15}}{6^5} = \frac{-9 \cdot 8 \cdot 7 \cdot 6}{1 \cdot 2 \cdot 3 \cdot 4} \cdot \frac{3^4}{2^5 \cdot 3^5} x^{19} = -\frac{21}{16} x^{19}$$

Ans.

(c) Term independent of x :

Term independent of x does not contain x ; Hence find the value of r for which the exponent of x is zero.

Illustration 7 : The term independent of x in $\left[\sqrt{\frac{x}{3}} + \sqrt{\left(\frac{3}{2x^2}\right)}\right]^{10}$ is –

- (A) 1 (B) $\frac{5}{12}$ (C) ${}^{10}C_1$ (D) none of these

Solution : General term in the expansion is

$${}^{10}C_r \left(\frac{x}{3}\right)^{\frac{r}{2}} \left(\frac{3}{2x^2}\right)^{\frac{10-r}{2}} = {}^{10}C_r x^{\frac{3r}{2}-10} \cdot \frac{3^{5-r}}{2^{\frac{10-r}{2}}} \text{ For constant term, } \frac{3r}{2} = 10 \Rightarrow r = \frac{20}{3}$$

which is not an integer. Therefore, there will be no constant term.

Ans. (D)

Do yourself - 2 :

- (i) Find the 7th term of $\left(3x^2 - \frac{1}{3}\right)^{10}$
- (ii) Find the term independent of x in the expansion : $\left(2x^2 - \frac{3}{x^3}\right)^{25}$
- (iii) Find the middle term in the expansion of : (a) $\left(\frac{2x}{3} - \frac{3}{2x}\right)^6$ (b) $\left(2x^2 - \frac{1}{x}\right)^7$

(d) Numerically greatest term :

Let numerically greatest term in the expansion of $(a + b)^n$ be T_{r+1} .

$$\Rightarrow \begin{cases} |T_{r+1}| \geq |T_r| \\ |T_{r+1}| \geq |T_{r+2}| \end{cases} \text{ where } T_{r+1} = {}^nC_r a^{n-r} b^r$$

Solving above inequalities we get $\frac{n+1}{1 + \left|\frac{a}{b}\right|} - 1 \leq r \leq \frac{n+1}{1 + \left|\frac{a}{b}\right|}$

Case I : When $\frac{n+1}{1 + \left|\frac{a}{b}\right|}$ is an integer equal to m, then T_m and T_{m+1} will be numerically greatest term.

Case II : When $\frac{n+1}{1 + \left|\frac{a}{b}\right|}$ is not an integer and its integral part is m, then T_{m+1} will be the numerically greatest term.

Illustration 8 : Find numerically greatest term in the expansion of $(3 - 5x)^{11}$ when $x = \frac{1}{5}$

Solution : Using $\frac{n+1}{1 + \left|\frac{a}{b}\right|} - 1 \leq r \leq \frac{n+1}{1 + \left|\frac{a}{b}\right|}$

$$\frac{11+1}{1 + \left|\frac{3}{-5x}\right|} - 1 \leq r \leq \frac{11+1}{1 + \left|\frac{3}{-5x}\right|}$$

solving we get $2 \leq r \leq 3$

$$\therefore r = 2, 3$$

so, the greatest terms are T_{2+1} and T_{3+1} .

$\therefore$ Greatest term (when $r = 2$)

$$T_3 = {}^{11}C_2 \cdot 3^9 (-5x)^2 = 55 \cdot 3^9 = T_4$$

From above we say that the value of both greatest terms are equal.

Ans.

Illustration 9 : Given T_3 in the expansion of $(1 - 3x)^6$ has maximum numerical value. Find the range of 'x'.

Solution : Using $\frac{n+1}{1+\left|\frac{a}{b}\right|} - 1 \leq r \leq \frac{n+1}{1+\left|\frac{a}{b}\right|}$

$$\frac{6+1}{1+\left|\frac{1}{-3x}\right|} - 1 \leq 2 \leq \frac{7}{1+\left|\frac{1}{-3x}\right|}$$

Let $|x| = t$

$$\frac{21t}{3t+1} - 1 \leq 2 \leq \frac{21t}{3t+1}$$

$$\begin{cases} \frac{21t}{3t+1} \leq 3 \\ \frac{21t}{3t+1} \geq 2 \end{cases} \Rightarrow \begin{cases} \frac{4t-1}{3t+1} \leq 0 \Rightarrow t \in \left[-\frac{1}{3}, \frac{1}{4}\right] \\ \frac{15t-2}{3t+1} \geq 0 \Rightarrow t \in \left(-\infty, -\frac{1}{3}\right] \cup \left[\frac{2}{15}, \infty\right) \end{cases}$$

$$\text{Common solution } t \in \left[\frac{2}{15}, \frac{1}{4}\right] \Rightarrow x \in \left[-\frac{1}{4}, -\frac{2}{15}\right] \cup \left[\frac{2}{15}, \frac{1}{4}\right]$$

Do yourself -3 :

- (i) Find the numerically greatest term in the expansion of $(3 - 2x)^9$, when $x = 1$.
- (ii) In the expansion of $\left(\frac{1}{2} + \frac{2x}{3}\right)^n$ when $x = -\frac{1}{2}$, it is known that 3rd terms is the greatest term. Find the possible integral values of n.

4. PROPERTIES OF BINOMIAL COEFFICIENTS :

$$(1 + x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n = \sum_{r=0}^n {}^nC_r x^r; n \in \mathbb{N} \quad \dots\dots(i)$$

where $C_0, C_1, C_2, \dots, C_n$ are called combinatorial (binomial) coefficients.

(a) The sum of all the binomial coefficients is 2^n .

Put $x = 1$, in (i) we get

$$C_0 + C_1 + C_2 + \dots + C_n = 2^n \Rightarrow \sum_{r=0}^n {}^nC_r = 2^n \quad \dots\dots(ii)$$

(b) Put $x = -1$ in (i) we get

$$C_0 - C_1 + C_2 - C_3 + \dots + C_n = 0 \Rightarrow \sum_{r=0}^n (-1)^r {}^nC_r = 0 \quad \dots\dots(iii)$$

(c) The sum of the binomial coefficients at odd position is equal to the sum of the binomial coefficient at even position and each is equal to 2^{n-1} .

$$\text{From (ii) \& (iii), } C_0 + C_2 + C_4 + \dots = C_1 + C_3 + C_5 + \dots = 2^{n-1}$$

(d) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

$$(e) \quad \frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$$

$$(f) \quad {}^nC_r = \frac{n}{r} \cdot {}^{n-1}C_{r-1} = \frac{n}{r} \cdot \frac{n-1}{r-1} \cdot {}^{n-2}C_{r-2} = \dots = \frac{n(n-1)(n-2)\dots(n-r+1)}{r(r-1)(r-2)\dots 1}$$

$$(g) \quad {}^nC_r = \frac{r+1}{n+1} \cdot {}^{n+1}C_{r+1}$$

Illustration 10 : Prove that : ${}^{25}C_{10} + {}^{24}C_{10} + \dots + {}^{10}C_{10} = {}^{26}C_{11}$

Solution : LHS = ${}^{10}C_{10} + {}^{11}C_{10} + {}^{12}C_{10} + \dots + {}^{25}C_{10}$
 $\Rightarrow {}^{11}C_{11} + {}^{11}C_{10} + {}^{12}C_{10} + \dots + {}^{25}C_{10}$
 $\Rightarrow {}^{12}C_{11} + {}^{12}C_{10} + \dots + {}^{25}C_{10}$
 $\Rightarrow {}^{13}C_{11} + {}^{13}C_{10} + \dots + {}^{25}C_{10}$
 and so on. $\therefore$ LHS = ${}^{26}C_{11}$

Alter :

LHS = coefficient of x^{10} in $\{(1+x)^{10} + (1+x)^{11} + \dots + (1+x)^{25}\}$
 $\Rightarrow$ coefficient of x^{10} in $\left[(1+x)^{10} \frac{(1+x)^{16}-1}{1+x-1}\right]$
 $\Rightarrow$ coefficient of x^{10} in $\frac{[(1+x)^{26} - (1+x)^{10}]}{x}$
 $\Rightarrow$ coefficient of x^{11} in $[(1+x)^{26} - (1+x)^{10}] = {}^{26}C_{11} - 0 = {}^{26}C_{11}$

Illustration 11 : A student is allowed to select at most n books from a collection of $(2n+1)$ books. If the total number of ways in which he can select books is 63, find the value of n .

Solution : Given student selects at most n books from a collection of $(2n+1)$ books. It means that he selects one book or two books or three books or or n books. Hence, by the given condition-

$${}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \dots + {}^{2n+1}C_n = 63 \quad \dots(i)$$

But we know that

$${}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \dots + {}^{2n+1}C_{2n+1} = 2^{2n+1} \quad \dots(ii)$$

Since ${}^{2n+1}C_0 = {}^{2n+1}C_{2n+1} = 1$, equation (ii) can also be written as

$$2 + ({}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \dots + {}^{2n+1}C_n) + ({}^{2n+1}C_{n+1} + {}^{2n+1}C_{n+2} + {}^{2n+1}C_{n+3} + \dots + {}^{2n+1}C_{2n-1} + {}^{2n+1}C_{2n}) = 2^{2n+1}$$

$$\Rightarrow 2 + ({}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \dots + {}^{2n+1}C_n) + ({}^{2n+1}C_n + {}^{2n+1}C_{n-1} + \dots + {}^{2n+1}C_2 + {}^{2n+1}C_1) = 2^{2n+1}$$

$$(\because {}^{2n+1}C_r = {}^{2n+1}C_{2n+1-r})$$

$$\Rightarrow 2 + 2({}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \dots + {}^{2n+1}C_n) = 2^{2n+1} \quad [\text{from (i)}]$$

$$\Rightarrow 2 + 2.63 = 2^{2n+1} \quad \Rightarrow \quad 1 + 63 = 2^{2n}$$

$$\Rightarrow 64 = 2^{2n} \Rightarrow 2^6 = 2^{2n}$$

$$\therefore \quad 2n = 6 \quad \text{Hence, } n = 3. \quad \text{Ans.}$$

Illustration 12 : Prove that :

$$(i) C_1 + 2C_2 + 3C_3 + \dots + nC_n = n \cdot 2^{n-1}$$

$$(ii) C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{n+1}$$

Solution :

$$(i) \text{ L.H.S.} = \sum_{r=1}^n r \cdot {}^nC_r = \sum_{r=1}^n r \cdot \frac{n}{r} \cdot {}^{n-1}C_{r-1} \\ = n \sum_{r=1}^n {}^{n-1}C_{r-1} = n \cdot [{}^{n-1}C_0 + {}^{n-1}C_1 + \dots + {}^{n-1}C_{n-1}] = n \cdot 2^{n-1}$$

Aliter : (Using method of differentiation)

$$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n \quad \dots\dots\dots(A)$$

Differentiating (A), we get

$$n(1+x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + \dots + n \cdot C_nx^{n-1}.$$

Put $x = 1$,

$$C_1 + 2C_2 + 3C_3 + \dots + n \cdot C_n = n \cdot 2^{n-1}$$

$$(ii) \text{ L.H.S.} = \sum_{r=0}^n \frac{C_r}{r+1} = \frac{1}{n+1} \sum_{r=0}^n \frac{n+1}{r+1} {}^nC_r \\ = \frac{1}{n+1} \sum_{r=0}^n {}^{n+1}C_{r+1} = \frac{1}{n+1} [{}^{n+1}C_1 + {}^{n+1}C_2 + \dots + {}^{n+1}C_{n+1}] = \frac{1}{n+1} [2^{n+1} - 1]$$

Aliter : (Using method of integration)

Integrating (A), we get

$$\frac{(1+x)^{n+1}}{n+1} + C = C_0x + \frac{C_1x^2}{2} + \frac{C_2x^3}{3} + \dots + \frac{C_nx^{n+1}}{n+1} \quad (\text{where } C \text{ is a constant})$$

$$\text{Put } x = 0, \text{ we get, } C = -\frac{1}{n+1}$$

$$\therefore \frac{(1+x)^{n+1} - 1}{n+1} = C_0x + \frac{C_1x^2}{2} + \frac{C_2x^3}{3} + \dots + \frac{C_nx^{n+1}}{n+1}$$

$$\text{Put } x = 1, \text{ we get } C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{n+1}$$

$$\text{Put } x = -1, \text{ we get } C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \dots + \frac{1}{n+1}$$

Illustration 13 : If $(1+x)^n = \sum_{r=0}^n {}^nC_r x^r$, then prove that $C_1^2 + 2 \cdot C_2^2 + 3 \cdot C_3^2 + \dots + n \cdot C_n^2 = \frac{(2n-1)!}{((n-1)!)^2}$

Solution :

$$(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n \quad \dots\dots\dots(i)$$

Differentiating both the sides, w.r.t. x , we get

$$n(1+x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + \dots + n \cdot C_nx^{n-1} \quad \dots\dots\dots(ii)$$

also, we have

$$(x+1)^n = C_0x^n + C_1x^{n-1} + C_2x^{n-2} + \dots + C_n \quad \dots\dots\dots(iii)$$

Multiplying (ii) & (iii), we get

$$(C_1 + 2C_2x + 3C_3x^2 + \dots + C_n x^{n-1})(C_0x^n + C_1x^{n-1} + C_2x^{n-2} + \dots + C_n) = n(1+x)^{2n-1}$$

Equating the coefficients of x^{n-1} , we get

$$C_1^2 + 2C_2^2 + 3C_3^2 + \dots + n.C_n^2 = n. {}^{2n-1}C_{n-1} = \frac{(2n-1)!}{((n-1)!)^2} \quad \text{Ans.}$$

Illustration 14 : Prove that : $C_0 - 3C_1 + 5C_2 - \dots - (-1)^n(2n+1)C_n = 0$

Solution :

$$T_r = (-1)^r(2r+1) {}^nC_r = 2(-1)^r r \cdot {}^nC_r + (-1)^m C_r$$

$$\Sigma T_r = 2 \sum_{r=1}^n (-1)^r \cdot r \cdot {}^{n-1}C_{r-1} + \sum_{r=0}^n (-1)^r {}^nC_r = 2 \sum_{r=1}^n (-1)^r \cdot {}^{n-1}C_{r-1} + \sum_{r=0}^n (-1)^r \cdot {}^nC_r$$

$$= 2 \left[{}^{n-1}C_0 - {}^{n-1}C_1 + \dots \right] + \left[{}^nC_0 - {}^nC_1 + \dots \right] = 0$$

Illustration 15 : Prove that $({}^{2n}C_0)^2 - ({}^{2n}C_1)^2 + ({}^{2n}C_2)^2 - \dots + (-1)^n ({}^{2n}C_{2n})^2 = (-1)^n \cdot {}^{2n}C_n$

Solution :

$$(1-x)^{2n} = {}^{2n}C_0 - {}^{2n}C_1x + {}^{2n}C_2x^2 - \dots + (-1)^{2n} {}^{2n}C_{2n}x^{2n} \quad \dots(i)$$

$$\text{and } (x+1)^{2n} = {}^{2n}C_0x^{2n} + {}^{2n}C_1x^{2n-1} + {}^{2n}C_2x^{2n-2} + \dots + {}^{2n}C_{2n} \quad \dots(ii)$$

Multiplying (i) and (ii), we get

$$(x^2-1)^{2n} = ({}^{2n}C_0 - {}^{2n}C_1x + \dots + (-1)^{n2n} {}^{2n}C_{2n}x^{2n}) \times ({}^{2n}C_0x^{2n} + {}^{2n}C_1x^{2n-1} + \dots + {}^{2n}C_{2n}) \quad \dots(iii)$$

Now, coefficient of x^{2n} in R.H.S.

$$= ({}^{2n}C_0)^2 - ({}^{2n}C_1)^2 + ({}^{2n}C_2)^2 - \dots + (-1)^n ({}^{2n}C_{2n})^2$$

$$\therefore \text{General term in L.H.S., } T_{r+1} = {}^{2n}C_r (x^2)^{2n-r} (-1)^r$$

Putting $2(2n-r) = 2n$

$$\therefore r = n$$

$$\therefore T_{n+1} = {}^{2n}C_n x^{2n} (-1)^n$$

Hence coefficient of x^{2n} in L.H.S. = $(-1)^n \cdot {}^{2n}C_n$

But (iii) is an identity, therefore coefficient of x^{2n} in R.H.S. = coefficient of x^{2n} in L.H.S.

$$\Rightarrow ({}^{2n}C_0)^2 - ({}^{2n}C_1)^2 + ({}^{2n}C_2)^2 - \dots + (-1)^n ({}^{2n}C_{2n})^2 = (-1)^n \cdot {}^{2n}C_n$$

Illustration 16 : Prove that : ${}^nC_0 \cdot {}^{2n}C_n - {}^nC_1 \cdot {}^{2n-2}C_{n-2} + {}^nC_2 \cdot {}^{2n-4}C_{n-4} + \dots = 2^n$

Solution :

$$\text{L.H.S.} = \text{Coefficient of } x^n \text{ in } [{}^nC_0(1+x)^{2n} - {}^nC_1(1+x)^{2n-2} + \dots]$$

$$= \text{Coefficient of } x^n \text{ in } [(1+x)^2 - 1]^n$$

$$= \text{Coefficient of } x^n \text{ in } x^n(x+2)^n = 2^n$$

Illustration 17 : If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$ then show that the sum of the products of

the C_i 's taken two at a time represented by : $\sum_{0 \leq i < j \leq n} C_i C_j$ is equal to $2^{2n-1} - \frac{2n!}{2.n!n!}$

Solution :

$$\text{Since } (C_0 + C_1 + C_2 + \dots + C_{n-1} + C_n)^2$$

$$= C_0^2 + C_1^2 + C_2^2 + \dots + C_{n-1}^2 + C_n^2 + 2(C_0C_1 + C_0C_2 + C_0C_3 + \dots + C_0C_n + C_1C_2$$

$$+ C_1C_3 + \dots + C_1C_n + C_2C_3 + C_2C_4 + \dots + C_2C_n + \dots + C_{n-1}C_n)$$

$$(2^n)^2 = {}^{2n}C_n + 2 \sum_{0 \leq i < j \leq n} C_i C_j$$

$$\text{Hence } \sum_{0 \leq i < j \leq n} C_i C_j = 2^{2n-1} - \frac{2n!}{2 \cdot n!n!}$$

Ans.

Illustration 18 : If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$ then prove that

$$\sum_{0 \leq i < j \leq n} (C_i + C_j)^2 = (n-1) {}^{2n}C_n + 2^{2n}$$

Solution :

$$\begin{aligned} \text{L.H.S. } & \sum_{0 \leq i < j \leq n} (C_i + C_j)^2 \\ &= (C_0 + C_1)^2 + (C_0 + C_2)^2 + \dots + (C_0 + C_n)^2 + (C_1 + C_2)^2 + (C_1 + C_3)^2 + \dots \\ & \quad + (C_1 + C_n)^2 + (C_2 + C_3)^2 + (C_2 + C_4)^2 + \dots + (C_2 + C_n)^2 + \dots + (C_{n-1} + C_n)^2 \\ &= n(C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2) + 2 \sum_{0 \leq i < j \leq n} C_i C_j \\ &= n \cdot {}^{2n}C_n + 2 \cdot \left\{ 2^{2n-1} - \frac{2n!}{2 \cdot n!n!} \right\} \quad \{\text{from Illustration 17}\} \\ &= n \cdot {}^{2n}C_n + 2^{2n} - {}^{2n}C_n = (n-1) \cdot {}^{2n}C_n + 2^{2n} = \text{R.H.S.} \end{aligned}$$

Do yourself - 4 :

(i) $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} =$

(A) 2^{n-1} (B) ${}^{2n}C_n$ (C) 2^n (D) 2^{n+1}

(ii) If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$, $n \in \mathbb{N}$. Prove that

(a) $3C_0 - 8C_1 + 13C_2 - 18C_3 + \dots$ upto $(n+1)$ terms $= 0$, if $n \geq 2$.

(b) $2C_0 + 2^2 \frac{C_1}{2} + 2^3 \frac{C_2}{3} + 2^4 \frac{C_3}{4} + \dots + 2^{n+1} \frac{C_n}{n+1} = \frac{3^{n+1} - 1}{n+1}$

(c) $C_0^2 + \frac{C_1^2}{2} + \frac{C_2^2}{3} + \dots + \frac{C_n^2}{n+1} = \frac{(2n+1)!}{((n+1)!)^2}$

5. MULTINOMIAL THEOREM :

Using binomial theorem, we have $(x+a)^n = \sum_{r=0}^n {}^nC_r x^{n-r} a^r$, $n \in \mathbb{N}$

$$= \sum_{r=0}^n \frac{n!}{(n-r)!r!} x^{n-r} a^r = \sum_{r+s=n} \frac{n!}{r!s!} x^s a^r, \text{ where } s+r=n$$

This result can be generalized in the following form.

$$(x_1 + x_2 + \dots + x_k)^n = \sum_{r_1+r_2+\dots+r_k=n} \frac{n!}{r_1!r_2!\dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k}$$

The general term in the above expansion $\frac{n!}{r_1!r_2!.....r_k!} \cdot x_1^{r_1} x_2^{r_2} x_3^{r_3} \dots x_k^{r_k}$

The number of terms in the above expansion is equal to the number of non-negative integral solution of the equation $r_1 + r_2 + \dots + r_k = n$ because each solution of this equation gives a term in the above expansion. The number of such solutions is ${}^{n+k-1}C_{k-1}$

Particular cases :

$$(i) \quad (x + y + z)^n = \sum_{r+s+t=n} \frac{n!}{r!s!t!} x^r y^s z^t$$

The above expansion has ${}^{n+3-1}C_{3-1} = {}^{n+2}C_2$ terms

$$(ii) \quad (x + y + z + u)^n = \sum_{p+q+r+s=n} \frac{n!}{p!q!r!s!} x^p y^q z^r u^s$$

There are ${}^{n+4-1}C_{4-1} = {}^{n+3}C_3$ terms in the above expansion.

Illustration 19 : Find the coefficient of $x^2 y^3 z^4 w$ in the expansion of $(x - y - z + w)^{10}$

Solution : $(x - y - z + w)^{10} = \sum_{p+q+r+s=10} \frac{n!}{p!q!r!s!} (x)^p (-y)^q (-z)^r (w)^s$

We want to get $x^2 y^3 z^4 w$ this implies that $p = 2, q = 3, r = 4, s = 1$

$\therefore$ Coefficient of $x^2 y^3 z^4 w$ is $\frac{10!}{2!3!4!1!} (-1)^3 (-1)^4 = -12600$ **Ans.**

Illustration 20 : Find the total number of terms in the expansion of $(1 + x + y)^{10}$ and coefficient of $x^2 y^3$.

Solution : Total number of terms = ${}^{10+3-1}C_{3-1} = {}^{12}C_2 = 66$

Coefficient of $x^2 y^3 = \frac{10!}{2! \times 3! \times 5!} = 2520$ **Ans.**

Illustration 21 : Find the coefficient of x^5 in the expansion of $(2 - x + 3x^2)^6$.

Solution : The general term in the expansion of $(2 - x + 3x^2)^6 = \frac{6!}{r!s!t!} 2^r (-x)^s (3x^2)^t$

where $r + s + t = 6$.

$$= \frac{6!}{r!s!t!} 2^r (-1)^s (3)^t x^{s+2t}$$

For the coefficient of x^5 , we must have $s + 2t = 5$.

But, $r + s + t = 6$,

$\therefore s = 5 - 2t$ and $r = 1 + t$, where $0 \leq r, s, t \leq 6$.

Now $t = 0 \Rightarrow r = 1, s = 5$.

$t = 1 \Rightarrow r = 2, s = 3$.

$t = 2 \Rightarrow r = 3, s = 1$.

Thus, there are three terms containing x^5 and coefficient of x^5

$$= \frac{6!}{1!5!0!} 2^1 \times (-1)^5 \times 3^0 + \frac{6!}{2!3!1!} \times 2^2 \times (-1)^3 \times 3^1 + \frac{6!}{3!1!2!} \times 2^3 \times (-1)^1 \times 3^2$$

$$= -12 - 720 - 4320 = -5052.$$

Ans.

Illustration 22 : If $(1 + x + x^2)^n = \sum_{r=0}^{2n} a_r x^r$, then prove that (a) $a_r = a_{2n-r}$ (b) $\sum_{r=0}^{n-1} a_r = \frac{1}{2} (3^n - a_n)$

Solution : (a) We have

$$(1 + x + x^2)^n = \sum_{r=0}^{2n} a_r x^r \quad \dots(A)$$

Replace x by $\frac{1}{x}$

$$\therefore \left(1 + \frac{1}{x} + \frac{1}{x^2}\right)^n = \sum_{r=0}^{2n} a_r \left(\frac{1}{x}\right)^r \Rightarrow (x^2 + x + 1)^n = \sum_{r=0}^{2n} a_r x^{2n-r}$$

$$\Rightarrow \sum_{r=0}^{2n} a_r x^r = \sum_{r=0}^{2n} a_r x^{2n-r} \quad \{\text{Using (A)}\}$$

Equating the coefficient of x^{2n-r} on both sides, we get
 $a_{2n-r} = a_r$ for $0 \leq r \leq 2n$.

Hence $a_r = a_{2n-r}$.

(b) Putting $x=1$ in given series, then

$$a_0 + a_1 + a_2 + \dots + a_{2n} = (1+1+1)^n$$

$$a_0 + a_1 + a_2 + \dots + a_{2n} = 3^n \quad \dots(1)$$

But $a_r = a_{2n-r}$ for $0 \leq r \leq 2n$

$\therefore$ series (1) reduces to

$$2(a_0 + a_1 + a_2 + \dots + a_{n-1}) + a_n = 3^n.$$

$$\therefore a_0 + a_1 + a_2 + \dots + a_{n-1} = \frac{1}{2} (3^n - a_n)$$

Do yourself - 5 :

(i) Find the coefficient of $x^2 y^5$ in the expansion of $(3 + 2x - y)^{10}$.

6. APPLICATION OF BINOMIAL THEOREM :

Illustration 23 : If $(6\sqrt{6} + 14)^{2n+1} = [N] + F$ and $F = N - [N]$; where $[.]$ denotes greatest integer function, then NF is equal to

- (A) 20^{2n+1} (B) an even integer (C) odd integer (D) 40^{2n+1}

Solution : Since $(6\sqrt{6} + 14)^{2n+1} = [N] + F$

Let us assume that $f = (6\sqrt{6} - 14)^{2n+1}$; where $0 < f < 1$.

$$\text{Now, } [N] + F - f = (6\sqrt{6} + 14)^{2n+1} - (6\sqrt{6} - 14)^{2n+1}$$

$$= 2 \left[{}^{2n+1}C_1 (6\sqrt{6})^{2n} (14) + {}^{2n+1}C_3 (6\sqrt{6})^{2n-2} (14)^3 + \dots \right]$$

$$\Rightarrow [N] + F - f = \text{even integer.}$$

Now $0 < F < 1$ and $0 < f < 1$

so $-1 < F - f < 1$ and $F - f$ is an integer so it can only be zero

$$\text{Thus } NF = (6\sqrt{6} + 14)^{2n+1} (6\sqrt{6} - 14)^{2n+1} = 20^{2n+1}$$

Ans. (A,B)

Illustration 24 : Find the last three digits in 11^{50} .

Solution : Expansion of $(10 + 1)^{50} = {}^{50}C_0 10^{50} + {}^{50}C_1 10^{49} + \dots + {}^{50}C_{48} 10^2 + {}^{50}C_{49} 10 + {}^{50}C_{50}$

$$= \underbrace{{}^{50}C_0 10^{50} + {}^{50}C_1 10^{49} + \dots + {}^{50}C_{47} 10^3}_{1000 \text{ k}} + 49 \times 25 \times 100 + 500 + 1$$

$$\Rightarrow 1000 K + 123001 \quad \Rightarrow \quad \text{Last 3 digits are 001.}$$

Illustration 25 : Prove that $2222^{5555} + 5555^{2222}$ is divisible by 7.

Solution : When 2222 is divided by 7 it leaves a remainder 3.

So adding & subtracting 3^{5555} , we get :

$$E = \underbrace{2222^{5555} - 3^{5555}}_{E_1} + \underbrace{3^{5555} + 5555^{2222}}_{E_2}$$

For E_1 : Now since $2222 - 3 = 2219$ is divisible by 7, therefore E_1 is divisible by 7

($\because x^n - a^n$ is divisible by $x - a$)

For E_2 : 5555 when divided by 7 leaves remainder 4.

So adding and subtracting 4^{2222} , we get :

$$E_2 = 3^{5555} + 4^{2222} + 5555^{2222} - 4^{2222}$$

$$= (243)^{1111} + (16)^{1111} + (5555)^{2222} - 4^{2222}$$

Again $(243)^{1111} + 16^{1111}$ and $(5555)^{2222} - 4^{2222}$ are divisible by 7

($\because x^n + a^n$ is divisible by $x + a$ when n is odd)

Hence $2222^{1111} + 5555^{2222}$ is divisible by 7.

Do yourself - 6 :

- (i) Prove that $5^{25} - 3^{25}$ is divisible by 2.
- (ii) Find the remainder when the number 9^{100} is divided by 8.
- (iii) Find last three digits in 19^{100} .
- (iv) Let $R = (8 + 3\sqrt{7})^{20}$ and $[.]$ denotes greatest integer function, then prove that :
 (a) $[R]$ is odd (b) $R - [R] = 1 - \frac{1}{(8 + 3\sqrt{7})^{20}}$
- (v) Find the digit at unit's place in the number $17^{1995} + 11^{1995} - 7^{1995}$.

7. BINOMIAL THEOREM FOR NEGATIVE OR FRACTIONAL INDICES :

If $n \in \mathbb{Q}$, then $(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \frac{n(n-1)(n-2)}{3!} x^3 + \dots \infty$ provided $|x| < 1$.

Note :

- (i) When the index n is a positive integer the number of terms in the expansion of $(1+x)^n$ is finite i.e. $(n+1)$ & the coefficient of successive terms are : ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$
- (ii) When the index is other than a positive integer such as negative integer or fraction, the number of terms in the expansion of $(1+x)^n$ is infinite and the symbol nC_r cannot be used to denote the coefficient of the general term.
- (iii) Following expansion should be remembered ($|x| < 1$).
 - (a) $(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \dots \infty$
 - (b) $(1-x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \dots \infty$
 - (c) $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots \infty$
 - (d) $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots \infty$
 - (e) $(1+x)^{-3} = 1 - 3x + 6x^2 - 10x^3 + \dots + \frac{(-1)^r(r+1)(r+2)}{2!}x^r + \dots$
 - (f) $(1-x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + \dots + \frac{(r+1)(r+2)}{2!}x^r + \dots$
- (iv) The expansions in ascending powers of x are only valid if x is 'small'. If x is large i.e. $|x| > 1$ then we may find it convenient to expand in powers of $1/x$, which then will be small.

8. APPROXIMATIONS :

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{1.2}x^2 + \frac{n(n-1)(n-2)}{1.2.3}x^3 + \dots$$

If $x < 1$, the terms of the above expansion go on decreasing and if x be very small, a stage may be reached when we may neglect the terms containing higher powers of x in the expansion. Thus, if x be so small that its square and higher powers may be neglected then $(1+x)^n = 1 + nx$, approximately. This is an approximate value of $(1+x)^n$

Illustration 26 : If x is so small such that its square and higher powers may be neglected then find the approximate value of $\frac{(1-3x)^{1/2} + (1-x)^{5/3}}{(4+x)^{1/2}}$

Solution :

$$\begin{aligned} \frac{(1-3x)^{1/2} + (1-x)^{5/3}}{(4+x)^{1/2}} &= \frac{1 - \frac{3}{2}x + 1 - \frac{5x}{3}}{2\left(1 + \frac{x}{4}\right)^{1/2}} = \frac{1}{2} \left(2 - \frac{19}{6}x\right) \left(1 + \frac{x}{4}\right)^{-1/2} \\ &= \frac{1}{2} \left(2 - \frac{19}{6}x\right) \left(1 - \frac{x}{8}\right) = \frac{1}{2} \left(2 - \frac{x}{4} - \frac{19}{6}x\right) = 1 - \frac{x}{8} - \frac{19}{12}x = 1 - \frac{41}{24}x \end{aligned}$$

Ans.

Illustration 27 : The value of cube root of 1001 up to five decimal places is –
(A) 10.03333 (B) 10.00333 (C) 10.00033 (D) none of these

Solution : $(1001)^{1/3} = (1000 + 1)^{1/3} = 10 \left(1 + \frac{1}{1000} \right)^{1/3} = 10 \left\{ 1 + \frac{1}{3} \cdot \frac{1}{1000} + \frac{1/3(1/3-1)}{2!} \frac{1}{1000^2} + \dots \right\}$
 $= 10 \{ 1 + 0.0003333 - 0.00000011 + \dots \} = 10.00333$ **Ans. (B)**

Illustration 28 : The sum of $1 + \frac{1}{4} + \frac{1.3}{4.8} + \frac{1.3.5}{4.8.12} + \dots \infty$ is –

- (A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\sqrt{3}$ (D) $2^{3/2}$

Solution : Comparing with $1 + nx + \frac{n(n-1)}{2!} x^2 + \dots$

$nx = 1/4$ (i)

and $\frac{n(n-1)x^2}{2!} = \frac{1.3}{4.8}$ or $\frac{nx(nx-x)}{2!} = \frac{3}{32}$

$\Rightarrow \frac{1}{4} \left(\frac{1}{4} - x \right) = \frac{3}{16}$ (by (i))

$\Rightarrow \left(\frac{1}{4} - x \right) = \frac{3}{4}$

$\Rightarrow x = \frac{1}{4} - \frac{3}{4} = -\frac{1}{2}$ (ii)

putting the value of x in (i)

$n(-1/2) = 1/4 \Rightarrow n = -1/2$

$\therefore$ sum of series $= (1+x)^n = (1-1/2)^{-1/2} = (1/2)^{-1/2} = \sqrt{2}$

Ans. (A)

9. EXPONENTIAL SERIES :

- (a) e is an irrational number lying between 2.7 and 2.8. Its value correct upto 10 places of decimal is 2.7182818284.
 (b) Logarithms to the base 'e' are known as the **Napier an** system, so named after Napier, their inventor. They are also called **Natural Logarithm**.

(c) $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty$; where x may be any real or complex number and

$$e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n$$

(d) $a^x = 1 + \frac{x}{1!} \ell na + \frac{x^2}{2!} \ell n^2 a + \frac{x^3}{3!} \ell n^3 a + \dots$, where $a > 0$

(e) $e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots \infty$

10. LOGARITHMIC SERIES :

$$(a) \quad \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \infty, \text{ where } -1 < x \leq 1$$

$$(b) \quad \ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} + \dots \infty, \text{ where } -1 \leq x < 1$$

Remember : (i) $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \infty = \ln 2$ (ii) $e^{\ln x} = x$; for all $x > 0$

(iii) $\ln 2 = 0.693$

(iv) $\ln 10 = 2.303$

ANSWERS FOR DO YOURSELF

1. (i) ${}^5C_0 x (3x^2)^5 + {}^5C_1 (3x^2)^4 \left(-\frac{x}{2}\right) + {}^5C_2 (3x^2)^3 \left(-\frac{x}{2}\right)^2 + {}^5C_3 (3x^2)^2 \left(-\frac{x}{2}\right)^3 + {}^5C_4 (3x^2)^1 \left(-\frac{x}{2}\right)^4 + {}^5C_5 \left(-\frac{x}{2}\right)^5$
 (ii) ${}^nC_0 y^n + {}^nC_1 y^{n-1} \cdot x + {}^nC_2 y^{n-2} \cdot x^2 + \dots + {}^nC_n \cdot x^n$
2. (i) $\frac{70}{3} x^8$; (ii) $\frac{25!}{10!5!} 2^{15} 3^{10}$; (iii) (a) -20; (b) $-560x^5, 280x^2$
3. (i) 4th & 5th i.e. 489888 (ii) $n = 4, 5, 6$
4. (i) C
5. (i) -272160 or $-{}^{10}C_5 \times {}^5C_2 \times 108$
6. (ii) 1 (iii) 801 (v) 1

EXERCISE # 1

[SINGLE CORRECT CHOICE TYPE]

- If the coefficients of x^7 & x^8 in the expansion of $\left[2 + \frac{x}{3}\right]^n$ are equal, then the value of n is :
(A) 15 (B) 45 (C) 55 (D) 56
- If the constant term of the binomial expansion $\left(2x - \frac{1}{x}\right)^n$ is -160 , then n is equal to -
(A) 4 (B) 6 (C) 8 (D) 10
- The coefficient of x^{49} in the expansion of $(x-1)\left(x - \frac{1}{2}\right)\left(x - \frac{1}{2^2}\right) \dots \left(x - \frac{1}{2^{49}}\right)$ is equal to -
(A) $-2\left(1 - \frac{1}{2^{50}}\right)$ (B) +ve coefficient of x
(C) -ve coefficient of x (D) $-2\left(1 - \frac{1}{2^{49}}\right)$
- Set of value of r for which, ${}^{18}C_{r-2} + 2 \cdot {}^{18}C_{r-1} + {}^{18}C_r \geq {}^{20}C_{13}$ contains :
(A) 4 element (B) 5 elements (C) 7 elements (D) 10 elements
- Number of rational terms in the expansion of $(\sqrt{2} + \sqrt[4]{3})^{100}$ is :
(A) 25 (B) 26 (C) 27 (D) 28
- The largest real value for x such that $\sum_{k=0}^4 \left(\frac{5^{4-k}}{(4-k)!}\right) \left(\frac{x^k}{k!}\right) = \frac{8}{3}$ is -
(A) $2\sqrt{2} - 5$ (B) $2\sqrt{2} + 5$ (C) $-2\sqrt{2} - 5$ (D) $-2\sqrt{2} + 5$
- The expression $[x + (x^3 - 1)^{1/2}]^5 + [x - (x^3 - 1)^{1/2}]^5$ is a polynomial of degree
(A) 5 (B) 6 (C) 7 (D) 8
- Given $(1 - 2x + 5x^2 - 10x^3)(1 + x)^n = 1 + a_1x + a_2x^2 + \dots$ and that $a_1^2 = 2a_2$ then the value of n is-
(A) 6 (B) 2 (C) 5 (D) 3
- The sum of the coefficients of all the even powers of x in the expansion of $(2x^2 - 3x + 1)^{11}$ is -
(A) 2.6^{10} (B) 3.6^{10} (C) 6^{11} (D) none

10. Co-efficient of α^t in the expansion of ,
 $(\alpha + p)^{m-1} + (\alpha + p)^{m-2}(\alpha + q) + (\alpha + p)^{m-3}(\alpha + q)^2 + \dots + (\alpha + q)^{m-1}$ where $\alpha \neq -q$ and $p \neq q$ is:
 (A) $\frac{{}^m C_t (p^t - q^t)}{p - q}$ (B) $\frac{{}^m C_t (p^{m-t} - q^{m-t})}{p - q}$ (C) $\frac{{}^m C_t (p^t + q^t)}{p - q}$ (D) $\frac{{}^m C_t (p^{m-t} + q^{m-t})}{p - q}$
11. If $n \in \mathbb{N}$ & n is even, then $\frac{1}{1!(n-1)!} + \frac{1}{3!(n-3)!} + \frac{1}{5!(n-5)!} + \dots + \frac{1}{(n-1)!1!} =$
 (A) 2^n (B) $\frac{2^{n-1}}{n!}$ (C) $2^n n!$ (D) none of these
12. Let $\binom{n}{k}$ represents the combination of 'n' things taken 'k' at a time, then the value of the sum
 $\binom{99}{97} + \binom{98}{96} + \binom{97}{95} + \dots + \binom{3}{1} + \binom{2}{0}$ equals -
 (A) $\binom{99}{97}$ (B) $\binom{100}{98}$ (C) $\binom{99}{98}$ (D) $\binom{100}{97}$

[COMPREHENSION TYPE]

Paragraph for question nos. 13 to 15

If $n \in \mathbb{N}$ and if $(1 + 4x + 4x^2)^n = \sum_{r=0}^{2n} a_r x^r$, where $a_0, a_1, a_2, \dots, a_{2n}$ are real numbers.

13. The value of $2 \sum_{r=0}^n a_{2r}$, is
 (A) $9^n - 1$ (B) $9^n + 1$ (C) $9^n - 2$ (D) $9^n + 2$
14. The value of $2 \sum_{r=1}^n a_{2r-1}$, is -
 (A) $9^n - 1$ (B) $9^n + 1$ (C) $9^n - 2$ (D) $9^n + 2$
15. The value of a_{2n-1} is -
 (A) 2^{2n} (B) $n \cdot 2^{2n}$ (C) $(n-1)2^{2n}$ (D) $(n+1)2^{2n}$

EXERCISE # 2

1. (a) If the coefficients of $(2r + 4)^{\text{th}}$, $(r - 2)^{\text{th}}$ terms in the expansion of $(1 + x)^{18}$ are equal, find r .
 (b) If the coefficients of the r^{th} , $(r + 1)^{\text{th}}$ & $(r + 2)^{\text{th}}$ terms in the expansion of $(1 + x)^{14}$ are in AP, find r .
 (c) If the coefficients of 2^{nd} , 3^{rd} & 4^{th} terms in the expansion of $(1 + x)^{2n}$ are in AP, show that $2n^2 - 9n + 7 = 0$.
2. Find the term independent of x in the expansion of (i) $\left[\sqrt{\frac{x}{3}} + \frac{\sqrt{3}}{2x^2}\right]^{10}$ (ii) $\left[\frac{1}{2}x^{1/3} + x^{-1/5}\right]^8$
3. Prove that the ratio of the coefficient of x^{10} in $(1 - x^2)^{10}$ & the term independent of x in $\left(x - \frac{2}{x}\right)^{10}$ is $1 : 32$.
4. Find the sum of the series $\sum_{r=0}^n (-1)^r \cdot {}^nC_r \left[\frac{1}{2^r} + \frac{3^r}{2^{2r}} + \frac{7^r}{2^{3r}} + \frac{15^r}{2^{4r}} + \dots \text{up to } m \text{ terms} \right]$
5. Find numerically greatest term in the expansion of :
 (i) $(2 + 3x)^9$ when $x = \frac{3}{2}$ (ii) $(3 - 5x)^{15}$ when $x = \frac{1}{5}$
6. Find the term independent of x in the expansion of $(1 + x + 2x^3) \left(\frac{3x^2}{2} - \frac{1}{3x} \right)^9$.
7. Let $(1 + x^2)^2 \cdot (1 + x)^n = \sum_{K=0}^{n+4} a_K \cdot x^K$. If a_1 , a_2 & a_3 are in AP, find n .
8. Let $f(x) = 1 - x + x^2 - x^3 + \dots + x^{16} - x^{17} = a_0 + a_1(1 + x) + a_2(1 + x)^2 + \dots + a_{17}(1 + x)^{17}$, find the value of a_2 .
9. Let $N = {}^{2000}C_1 + 2 \cdot {}^{2000}C_2 + 3 \cdot {}^{2000}C_3 + \dots + 2000 \cdot {}^{2000}C_{2000}$. Prove that N is divisible by 2^{2003} .
10. Find the coefficient of
 (a) $x^2 y^3 z^4$ in the expansion of $(ax - by + cz)^9$.
 (b) $a^2 b^3 c^4 d$ in the expansion of $(a - b - c + d)^{10}$.
11. Find the coefficient of
 (a) x^4 in the expansion of $(1 + x + x^2 + x^3)^{11}$
 (b) x^4 in the expansion of $(2 - x + 3x^2)^6$
12. Find the coefficient of x^r in the expression :
 $(x + 3)^{n-1} + (x + 3)^{n-2}(x + 2) + (x + 3)^{n-3}(x + 2)^2 + \dots + (x + 2)^{n-1}$

13. (a) Show that the integral part in each of the following is odd. $n \in \mathbb{N}$
 (A) $(5 + 2\sqrt{6})^n$ (B) $(8 + 3\sqrt{7})^n$
 (b) Show that the integral part in each of the following is even. $n \in \mathbb{N}$
 (A) $(3\sqrt{3} + 5)^{2n+1}$ (B) $(5\sqrt{5} + 11)^{2n+1}$
14. Given that $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$, find the values of :
 (i) $a_0 + a_1 + a_2 + \dots + a_{2n}$; (ii) $a_0 - a_1 + a_2 - a_3 + \dots + a_{2n}$;
 (iii) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + a_{2n}^2$
15. Prove the following identities using the theory of permutation where $C_0, C_1, C_2, \dots, C_n$ are the combinatorial coefficients in the expansion of $(1 + x)^n$, $n \in \mathbb{N}$:
 (a) $C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2 = \frac{(2n)!}{n!n!}$
 (b) $C_0 C_1 + C_1 C_2 + C_2 C_3 + \dots + C_{n-1} C_n = \frac{(2n)!}{(n+1)!(n-1)!}$
 (c) $C_0 C_r + C_1 C_{r+1} + C_2 C_{r+2} + \dots + C_{n-r} C_n = \frac{2n!}{(n-r)!(n+r)!}$
 (d) $\sum_{r=0}^{n-2} ({}^n C_r \cdot {}^n C_{r+2}) = \frac{(2n)!}{(n-2)!(n+2)!}$
 (e) ${}^{100}C_{10} + 5 \cdot {}^{100}C_{11} + 10 \cdot {}^{100}C_{12} + 10 \cdot {}^{100}C_{13} + 5 \cdot {}^{100}C_{14} + {}^{100}C_{15} = {}^{105}C_{90}$
16. If $C_0, C_1, C_2, \dots, C_n$ are the combinatorial coefficients in the expansion of $(1 + x)^n$, $n \in \mathbb{N}$, then prove the following :
 (a) $C_1 + 2C_2 + 3C_3 + \dots + n \cdot C_n = n \cdot 2^{n-1}$
 (b) $C_0 + 2C_1 + 3C_2 + \dots + (n+1)C_n = (n+2)2^{n-1}$
 (c) $C_0 + 3C_1 + 5C_2 + \dots + (2n+1)C_n = (n+1)2^n$
 (d) $(C_0 + C_1)(C_1 + C_2)(C_2 + C_3) \dots (C_{n-1} + C_n) = \frac{C_0 \cdot C_1 \cdot C_2 \dots C_{n-1} (n+1)^n}{n!}$
 (e) $1 \cdot C_0^2 + 3 \cdot C_1^2 + 5 \cdot C_2^2 + \dots + (2n+1) C_n^2 = \frac{(n+1)(2n)!}{n!n!}$
17. Prove that
 (a) $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + \dots + \frac{n \cdot C_n}{C_{n-1}} = \frac{n(n+1)}{2}$
 (b) $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{n+1}$
 (c) $2 \cdot C_0 + \frac{2^2 \cdot C_1}{2} + \frac{2^3 \cdot C_2}{3} + \frac{2^4 \cdot C_3}{4} + \dots + \frac{2^{n+1} \cdot C_n}{n+1} = \frac{3^{n+1} - 1}{n+1}$
 (d) $C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \dots + (-1)^n \frac{C_n}{n+1} = \frac{1}{n+1}$

EXERCISE # 3 (JM)

- For natural numbers m, n if $(1-y)^m (1+y)^n = 1 + a_1 y + a_2 y^2 + \dots$, and $a_1 = a_2 = 10$, then (m, n) is- **[AIEEE 2006]**
 (1) (45, 35) (2) (35, 45) (3) (20, 45) (4) (35, 20)
- The sum of the series ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \dots - \dots + {}^{20}C_{10}$ is- **[AIEEE 2007]**
 (1) $\frac{1}{2} {}^{20}C_{10}$ (2) 0 (3) $- {}^{20}C_{10}$ (4) ${}^{20}C_{10}$
- In the binomial expansion of $(a-b)^n$, $n \geq 5$, the sum of 5th and 6th terms is zero, then $\frac{a}{b}$ equals **[AIEEE 2007, IIT-2001]**
 (1) $\frac{6}{n-5}$ (2) $\frac{n-5}{6}$ (3) $\frac{n-4}{5}$ (4) $\frac{5}{n-4}$
- Statement-1:** $\sum_{r=0}^n (r+1) {}^nC_r = (n+2)2^{n-1}$
Statement-2: $\sum_{r=0}^n (r+1) {}^nC_r x^r = (1+x)^n + nx(1+x)^{n-1}$ **[AIEEE 2008]**
 (1) Statement-1 is false, Statement-2 is true.
 (2) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1
 (3) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1
 (4) Statement-1 is true, Statement-2 is false
- The remainder left out when $8^{2n} - (62)^{2n+1}$ is divided by 9 is :- **[AIEEE 2009]**
 (1) 7 (2) 8 (3) 0 (4) 2
- Let $S_1 = \sum_{j=1}^{10} j(j-1) {}^{10}C_j$, $S_2 = \sum_{j=1}^{10} j {}^{10}C_j$ and $S_3 = \sum_{j=1}^{10} j^2 {}^{10}C_j$. **[AIEEE 2010]**
Statement-1: $S_3 = 55 \times 2^9$.
Statement-2: $S_1 = 90 \times 2^8$ and $S_2 = 10 \times 2^8$.
 (1) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement-1.
 (2) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for Statement-1.
 (3) Statement-1 is true, Statement-2 is false.
 (4) Statement-1 is false, Statement-2 is true.
- The coefficient of x^7 in the expansion of $(1-x-x^2+x^3)^6$ is :- **[AIEEE 2011]**
 (1) -144 (2) 132 (3) 144 (4) -132

8. If n is a positive integer, then $(\sqrt{3} + 1)^{2n} - (\sqrt{3} - 1)^{2n}$ is : **[AIEEE 2012]**
 (1) a rational number other than positive integers
 (2) an irrational number
 (3) an odd positive integer
 (4) an even positive integer
9. The term independent of x in expansion of $\left(\frac{x+1}{x^{2/3} - x^{1/3} + 1} - \frac{x-1}{x - x^{1/2}}\right)^{10}$ is : **[JEE-Main 2013]**
 (1) 4 (2) 120 (3) 210 (4) 310
10. If the coefficients of x^3 and x^4 in the expansion of $(1 + ax + bx^2)(1 - 2x)^{18}$ in powers of x are both zero, then (a, b) is equal to :- **[JEE(Main)-2014]**
 (1) $\left(16, \frac{251}{3}\right)$ (2) $\left(14, \frac{251}{3}\right)$ (3) $\left(14, \frac{272}{3}\right)$ (4) $\left(16, \frac{272}{3}\right)$
11. The sum of coefficients of integral powers of x in the binomial expansion of $(1 - 2\sqrt{x})^{50}$ is : **[JEE(Main)-2015]**
 (1) $\frac{1}{2}(3^{50} - 1)$ (2) $\frac{1}{2}(2^{50} + 1)$ (3) $\frac{1}{2}(3^{50} + 1)$ (4) $\frac{1}{2}(3^{50})$
12. If the number of terms in the expansion of $\left(1 - \frac{2}{x} + \frac{4}{x^2}\right)^n$, $x \neq 0$, is 28, then the sum of the coefficients of all terms in this expansion is :- **[JEE(Main)-2016]**
 (1) 729 (2) 64 (3) 2187 (4) 243
13. The value of $({}^{21}C_1 - {}^{10}C_1) + ({}^{21}C_2 - {}^{10}C_2) + ({}^{21}C_3 - {}^{10}C_3) + ({}^{21}C_4 - {}^{10}C_4) + \dots + ({}^{21}C_{10} - {}^{10}C_{10})$ is : **[JEE(Main)-2017]**
 (1) $2^{20} - 2^{10}$ (2) $2^{21} - 2^{11}$ (3) $2^{21} - 2^{10}$ (4) $2^{20} - 2^9$
14. The sum of the co-efficient of all odd degree terms in the expansion of $(x + \sqrt{x^3 - 1})^5 + (x - \sqrt{x^3 - 1})^5$, $(x > 1)$ is : **[JEE(Main)-2018]**
 (1) 2 (2) -1 (3) 0 (4) 1
15. If the fractional part of the number $\frac{2^{403}}{15}$ is $\frac{k}{15}$, then k is equal to : **[JEE(Main) Online - 2019]**
 (1) 14 (2) 6 (3) 4 (4) 8

16. The coefficient of t^4 in the expansion of $\left(\frac{1-t^6}{1-t}\right)^3$ is : [JEE(Main) Online - 2019]
 (1) 12 (2) 10 (3) 15 (4) 14
17. If the third term in the binomial expansion of $(1+x^{\log_2 x})^5$ equals 2560, then a possible value of x is: [JEE(Main) Online - 2019]
 (1) $2\sqrt{2}$ (2) $4\sqrt{2}$ (3) $\frac{1}{8}$ (4) $\frac{1}{4}$
18. If $\sum_{i=1}^{20} \left(\frac{{}^{20}C_{i-1}}{{}^{20}C_i + {}^{20}C_{i-1}} \right)^3 = \frac{k}{21}$, then k equals: [JEE(Main) Online - 2019]
 (1) 200 (2) 100 (3) 50 (4) 400
19. The positive value of λ for which the co-efficient of x^2 in the expression $x^2 \left(\sqrt{x} + \frac{\lambda}{x^2} \right)^{10}$ is 720, is: [JEE(Main) Online - 2019]
 (1) 3 (2) $2\sqrt{2}$ (3) $\sqrt{5}$ (4) 4
20. If $\sum_{r=0}^{25} \{ {}^{50}C_r \cdot {}^{50-r}C_{25-r} \} = K({}^{50}C_{25})$, then K is equal to: [JEE(Main) Online - 2019]
 (1) 2^{25} (2) $2^{25} - 1$ (3) 2^{24} (4) $(25)^2$
21. The value of r for which ${}^{20}C_r \cdot {}^{20}C_0 + {}^{20}C_{r-1} \cdot {}^{20}C_1 + {}^{20}C_{r-2} \cdot {}^{20}C_2 + \dots + {}^{20}C_0 \cdot {}^{20}C_r$ is maximum, is : [JEE(Main) Online - 2019]
 (1) 15 (2) 10 (3) 20 (4) 11
22. The sum of the real values of x for which the middle term in binomial expansion of $\left(\frac{x^3}{3} + \frac{3}{x} \right)^8$ equals 5670 is : [JEE(Main) Online - 2019]
 (1) 8 (2) 0 (3) 4 (4) 6
23. Let $(x+10)^{50} + (x-10)^{50} = a_0 + a_1x + a_2x^2 + \dots + a_{50}x^{50}$, for all $x \in \mathbb{R}$; then $\frac{a_2}{a_0}$ is equal to: [JEE(Main) Online - 2019]
 (1) 12.00 (2) 12.75 (3) 12.25 (4) 12.50
24. Let $S_n = 1 + q + q^2 + \dots + q^n$ and $T_n = 1 + \left(\frac{q+1}{2}\right) + \left(\frac{q+1}{2}\right)^2 + \dots + \left(\frac{q+1}{2}\right)^n$ where q is a real number and $q \neq 1$. If ${}^{101}C_1 + {}^{101}C_2 \cdot S_1 + \dots + {}^{101}C_{101} \cdot S_{100} = \alpha T_{100}$, then α is equal to [JEE(Main) Online - 2019]
 (1) 200 (2) 2^{99} (3) 202 (4) 2^{100}

25. A ratio of the 5th term from the beginning to the 5th term the end in the binomial expansion of

$$\left(2^{1/3} + \frac{1}{2(3)^{1/3}} \right)^{10} \text{ is :}$$

[JEE(Main) Online - 2019]

- (1) $2(36)^{1/3} : 1$ (2) $1 : 4(16)^{1/3}$ (3) $1 : 2(16)^{1/3}$ (4) $4(36)^{1/3} : 1$

26. If nC_4 , nC_5 and nC_6 are in A.P., the n can be:

[JEE(Main) Online - 2019]

- (1) 11 (2) 12 (3) 9 (4) 14

27. The sum of the co-efficient of all even degree terms in x in the expansion of $(x + \sqrt{x^3 - 1})^6 + (x - \sqrt{x^3 - 1})^6$, ($x > 1$) is equal to

[JEE(Main) Online - 2019]

- (1) 26 (2) 24 (3) 29 (4) 32

28. If the fourth term in the binomial expansion of $\left(\sqrt{\frac{1}{x^{1+\log_{10}x}}} + x^{\frac{1}{12}} \right)^6$ is equal to 200, and $x > 1$, then the value of x is :

[JEE(Main) Online - 2019]

- (1) 100 (2) 10 (3) 10^4 (4) 10^3

***Note : The correct question is $\left(\sqrt{\frac{1}{x^{1+\log_{10}x}}} + x^{\frac{1}{12}} \right)^6$ & the solution is $x = 10$.**

29. If the fourth term in the Binomial expansion of $\left(\frac{2}{x} + x^{\log_8 x} \right)^6$ ($x > 0$) is 20×8^7 , then a value of x is:

[JEE(Main) Online - 2019]

- (1) 8^2 (2) 8 (3) 8^{-2} (4) 8^3

30. If some three consecutive coefficients in the binomial expansion of $(x + 1)^n$ in powers of x are in the ratio 2 : 15 : 70, then the average of these three coefficients is :

- (1) 232 (2) 625 (3) 964 (4) 227

[JEE(Main) Online - 2019]

31. If the coefficients of x^2 and x^3 are both zero, in the expansion of the expression $(1 + ax + bx^2)(1 - 3x)^{15}$ in powers of x, then the ordered pair (a, b) is equal to:

- (1) (28, 315) (2) (-21, 714) (3) (-54, 315) (4) (28, 861)

[JEE(Main) Online - 2019]

32. The smallest natural number n, such that the coefficient of x in the expansion of $\left(x^2 + \frac{1}{x^3} \right)^n$ is ${}^nC_{23}$, is :

[JEE(Main) Online - 2019]

- (1) 38 (2) 35 (3) 58 (4) 23

33. The coefficient of x^{18} in the product $(1+x)(1-x)^{10}(1+x+x^2)^9$ is
 (1) -84 (2) 126 (3) -126 (4) 84
[JEE(Main) Online - 2019]
34. If ${}^{20}C_1 + (2^2) {}^{20}C_2 + (3^2) {}^{20}C_3 + \dots + (20^2) {}^{20}C_{20} = A(2^\beta)$, then the ordered pair (A, β) is equal to:
 (1) (380, 18) (2) (380, 19) (3) (420, 18) (4) (420, 19)
[JEE(Main) Online - 2019]
35. The term independent of x in the expansion of $\left(\frac{1}{60} - \frac{x^8}{81}\right) \cdot \left(2x^2 - \frac{3}{x^2}\right)^6$ is equal to :
 (1) -108 (2) -72 (3) -36 (4) 36
[JEE(Main) Online - 2019]
36. If the sum of the coefficients of all even powers of x in the product $(1+x+x^2+\dots+x^{2n})(1-x+x^2-x^3+\dots+x^{2n})$ is 61, then n is equal to _____.
[JEE(Main)-2020]
37. The coefficient of x^7 in the expression $(1+x)^{10} + x(1+x)^9 + x^2(1+x)^8 + \dots + x^{10}$ is :
 (1) 420 (2) 330 (3) 120 (4) 210
[JEE(Main)-2020]
38. The number of ordered pairs (r, k) for which $6 \cdot {}^{35}C_r = (k^2 - 3) \cdot {}^{36}C_{r+1}$, where k is an integer is :
 (1) 2 (2) 4 (3) 3 (4) 6 **[JEE(Main)-2020]**
39. If a, b and c are the greatest values of ${}^{19}C_p, {}^{20}C_q$ and ${}^{21}C_r$ respectively, then :
 (1) $\frac{a}{11} = \frac{b}{22} = \frac{c}{21}$ (2) $\frac{a}{10} = \frac{b}{11} = \frac{c}{21}$ (3) $\frac{a}{10} = \frac{b}{11} = \frac{c}{42}$ (4) $\frac{a}{11} = \frac{b}{22} = \frac{c}{42}$
[JEE(Main)-2020]
40. If α and β be the coefficients of x^4 and x^2 respectively in the expansion of $(x + \sqrt{x^2 - 1})^6 + (x - \sqrt{x^2 - 1})^6$ then
 (1) $\alpha - \beta = -132$ (2) $\alpha + \beta = -30$ (3) $\alpha + \beta = 60$ (4) $\alpha - \beta = 60$
[JEE(Main)-2020]
41. The coefficient of x^4 in the expansion of $(1+x+x^2)^{10}$ is **[JEE(Main)-2020]**
42. In the expression of $\left(\frac{x}{\cos \theta} + \frac{1}{x \sin \theta}\right)^{16}$, if l_1 is the least value of term independent of x when $\frac{\pi}{8} \leq \theta \leq \frac{\pi}{4}$ and l_2 is the least value of the term independent of x when $\frac{\pi}{16} \leq \theta \leq \frac{\pi}{8}$, then the ratio $l_2 : l_1$ is equal to
 (1) 1 : 8 (2) 1 : 16 (3) 16 : 1 (4) 8 : 1
[JEE(Main)-2020]
43. If $C_r = {}^{25}C_r$ and $C_0 + 5 \cdot C_1 + 9 \cdot C_2 + \dots + (101) \cdot C_{25} = 2^{25} \cdot k$, then k is equal to _____.
[JEE(Main)-2020]

44. Let $\alpha > 0, \beta > 0$ be such that $\alpha^3 + \beta^2 = 4$. If the maximum value of the term independent of x in the binomial expansion of $(\alpha x^{1/9} + \beta x^{-1/6})^{10}$ is $10k$, then k is equal to : **[JEE(Main)-2020]**
(1) 352 (2) 336 (3) 176 (4) 84
45. If the number of integral terms in the expansion of $(3^{1/2} + 5^{1/8})^n$ is exactly 33, then the least value of n is : **[JEE(Main)-2020]**
(1) 128 (2) 264 (3) 256 (4) 248
46. The value of $(2 \cdot {}^1P_0 - 3 \cdot {}^2P_1 + 4 \cdot {}^3P_2 - \dots$ up to 51^{th} term) $+ (1! - 2! + 3! - \dots$ up to 51^{th} term) is equal to : **[JEE(Main)-2020]**
(1) $1 - 51(51)!$ (2) 1 (3) $1 + (51)!$ (4) $1 + (52)!$
47. If the term independent of x in the expansion of $\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ is k , then $18k$ is equal to : **[JEE(Main)-2020]**
(1) 7 (2) 9 (3) 5 (4) 11
48. The value of $\sum_{r=0}^{20} {}^{50-r}C_6$ is equal to : **[JEE(Main)-2020]**
(1) ${}^{50}C_6 - {}^{30}C_6$ (2) ${}^{51}C_7 + {}^{30}C_7$ (3) ${}^{51}C_7 - {}^{30}C_7$ (4) ${}^{50}C_7 - {}^{30}C_7$
49. Let $(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r$. Then $\frac{a_7}{a_{13}}$ is equal to _____. **[JEE(Main)-2020]**
50. If for some positive integer n , the coefficients of three consecutive terms in the binomial expansion of $(1+x)^{n+5}$ are in the ratio $5 : 10 : 14$, then the largest coefficient in this expansion is : **[JEE(Main)-2020]**
(1) 792 (2) 252 (3) 462 (4) 330
51. The natural number m , for which the coefficient of x in the binomial expansion of $\left(x^m + \frac{1}{x^2}\right)^{22}$ is 1540, is _____. **[JEE(Main)-2020]**
52. The coefficient of x^4 in the expansion of $(1+x+x^2+x^3)^6$ in powers of x , is _____. **[JEE(Main)-2020]**
53. If $\{p\}$ denotes the fractional part of the number p , then $\left\{\frac{3^{200}}{8}\right\}$, is equal to **[B.T. (E)]**
(1) $\frac{1}{8}$ (2) $\frac{5}{8}$ (3) $\frac{3}{8}$ (4) $\frac{7}{8}$ **[JEE(Main)-2020]**
54. If the constant term in the binomial expansion of $\left(\sqrt{x} - \frac{k}{x^2}\right)^{10}$ is 405, then $|k|$ equals : **[JEE(Main)-2020]**
(1) 2 (2) 1 (3) 3 (4) 9

EXERCISE # 4 (JA)

- The sum $\sum_{i=0}^m \binom{10}{i} \binom{20}{m-i}$, (where $\binom{p}{q} = 0$ if $p < q$) is maximum when m is
(A) 5 (B) 10 (C) 15 (D) 20
[JEE 2002 (Screening), 3]
- Coefficient of t^{24} in the expansion of $(1+t^2)^{12} (1+t^{12}) (1+t^{24})$ is
(A) $^{12}C_6 + 2$ (B) $^{12}C_6 + 1$ (C) $^{12}C_6$ (D) none
[JEE 2003, (Screening), 3]
- Prove that : $2^K \cdot \binom{n}{0} \binom{n}{K} - 2^{K-1} \binom{n}{1} \binom{n-1}{K-1} + 2^{K-2} \binom{n}{2} \binom{n-2}{K-2} \dots + (-1)^K \binom{n}{K} \binom{n-K}{0} = \binom{n}{K}$.
[JEE 2003, (Mains), 2]
- $^{n-1}C_r = (K^2 - 3) \cdot ^nC_{r+1}$, if $K \in$
(A) $[-\sqrt{3}, \sqrt{3}]$ (B) $(-\infty, -2)$ (C) $(2, \infty)$ (D) $(\sqrt{3}, 2]$
[JEE 2004 (Screening)]
- The value of $\binom{30}{0} \binom{30}{10} - \binom{30}{1} \binom{30}{11} + \binom{30}{2} \binom{30}{12} \dots + \binom{30}{20} \binom{30}{30}$ is where $\binom{n}{r} = ^nC_r$.
(A) $\binom{30}{10}$ (B) $\binom{30}{15}$ (C) $\binom{60}{30}$ (D) $\binom{31}{10}$
[JEE 2005 (Screening)]
- For $r = 0, 1, \dots, 10$, let A_r , B_r and C_r denote, respectively, the coefficient of x^r in the expansions of $(1+x)^{10}$, $(1+x)^{20}$ and $(1+x)^{30}$. Then $\sum_{r=1}^{10} A_r (B_{10} B_r - C_{10} A_r)$ is equal to -
(A) $B_{10} - C_{10}$ (B) $A_{10} (B_{10}^2 - C_{10} A_{10})$
(C) 0 (D) $C_{10} - B_{10}$
[JEE 2010, 5]
- The coefficients of three consecutive terms of $(1+x)^{n+5}$ are in the ratio 5 : 10 : 14. Then $n =$
[JEE (Advanced) 2013, 4M, 1M]
- Coefficient of x^{11} in the expansion of $(1+x^2)^4 (1+x^3)^7 (1+x^4)^{12}$ is -
(A) 1051 (B) 1106 (C) 1113 (D) 1120
[JEE(Adv.)-2014, 3(-1)]

9. The coefficient of x^9 in the expansion of $(1+x)(1+x^2)(1+x^3)\dots(1+x^{100})$ is
[JEE 2015, 4M, -0M]

10. Let m be the smallest positive integer such that the coefficient of x^2 in the expansion of $(1+x)^2 + (1+x)^3 + \dots + (1+x)^{49} + (1+mx)^{50}$ is $(3n+1)^{51}C_3$ for some positive integer n . Then the value of n is
[JEE(Advanced)-2016, 3(0)]

11. Let $X = \binom{10}{1}C_1^2 + 2\binom{10}{2}C_2^2 + 3\binom{10}{3}C_3^2 + \dots + 10\binom{10}{10}C_{10}^2$ where ${}^{10}C_r$, $r \in \{1, 2, \dots, 10\}$ denote binomial coefficients. Then, the value of $\frac{1}{1430}X$ is _____.
[JEE(Advanced)-2018, 3(0)]

12. Suppose $\det \begin{bmatrix} \sum_{k=0}^n k & \sum_{k=0}^n {}^nC_k k^2 \\ \sum_{k=0}^n {}^nC_k k & \sum_{k=0}^n {}^nC_k 3^k \end{bmatrix} = 0$
holds for some positive integer n . Then $\sum_{k=0}^n \frac{{}^nC_k}{k+1}$ equals _____.
[JEE(Advanced)-2019]

13. For non-negative integers s and r , let
[JEE(Advanced)-2020]

$$\binom{s}{r} = \begin{cases} \frac{s!}{r!(s-r)!} & \text{if } r \leq s, \\ 0 & \text{if } r > s. \end{cases}$$

For positive integers m and n , let

$$g(m, n) = \sum_{p=0}^{m+n} \frac{f(m, n, p)}{\binom{n+p}{p}}$$

where for any nonnegative integer p ,

$$f(m, n, p) = \sum_{i=0}^p \binom{m}{i} \binom{n+i}{p} \binom{p+n}{p-i}$$

Then which of the following statements is/are TRUE ?

- (A) $g(m, n) = g(n, m)$ for all positive integers m, n
- (B) $g(m, n+1) = g(m+1, n)$ for all positive integers m, n
- (C) $g(2m, 2n) = 2g(m, n)$ for all positive integer m, n
- (D) $g(2m, 2n) = (g(m, n))^2$ for all positive integers m, n

EXERCISE # 5

[SINGLE CORRECT CHOICE TYPE]

- Let $(5 + 2\sqrt{6})^n = p + f$ where $n, p \in \mathbb{N}$ and $0 < f < 1$ then the value of $f^2 - f + pf - p$ is –
(A) a natural number (B) a negative integer (C) a prime number (D) are irrational number
- Greatest term in the binomial expansion of $(a + 2x)^9$ when $a = 1$ & $x = \frac{1}{3}$ is :
(A) 3rd & 4th (B) 4th & 5th (C) only 4th (D) only 5th
- If $\sum_{r=1}^{10} r(r-1) {}^{10}C_r = k \cdot 2^9$, then k is equal to-
(A) 10 (B) 45 (C) 90 (D) 100
- The sum $\frac{\binom{11}{0}}{1} + \frac{\binom{11}{1}}{2} + \frac{\binom{11}{2}}{3} + \dots + \frac{\binom{11}{11}}{12}$ equals $\left(\text{where } \binom{n}{r} \text{ denotes } {}^nC_r \right)$
(A) $\frac{2^{11}}{12}$ (B) $\frac{2^{12}}{12}$ (C) $\frac{2^{11}-1}{12}$ (D) $\frac{2^{12}-1}{12}$
- Statement-1** :The sum of the series ${}^nC_0 \cdot {}^mC_r + {}^nC_1 \cdot {}^mC_{r-1} + {}^nC_2 \cdot {}^mC_{r-2} + \dots + {}^nC_r \cdot {}^mC_0$ is equal to ${}^{n+m}C_r$, where nC_i 's and mC_i 's denotes the combinatorial coefficients in the expansion of $(1+x)^n$ and $(1+x)^m$ respectively.
Statement-2 :Number of ways in which r children can be selected out of (n + m) children consisting of n boys and m girls if each selection may consist of any number of boys and girls is equal to ${}^{n+m}C_r$.
(A) Statement-1 is true, statement-2 is true ; statement-2 is a correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true ; statement-2 is NOT a correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

[MULTIPLE CORRECT CHOICE TYPE]

- In the expansion of $\left(\frac{x+1}{x^{2/3} - x^{1/3} + 1} - \frac{x-1}{x - x^{1/2}} \right)^{10}$, the term which does contain x^0 is -
(A) ${}^{11}C_4 - {}^{10}C_3$ (B) ${}^{10}C_7$ (C) ${}^{10}C_4$ (D) ${}^{11}C_5 - {}^{10}C_5$
- Let $(1 + x^2)^2 (1 + x)^n = A_0 + A_1x + A_2x^2 + \dots$. If A_0, A_1, A_2 are in A.P. then the value of n is-
(A) 2 (B) 3 (C) 5 (D) 7
- In the expansion of $\left(x^3 + 3 \cdot 2^{-\log_2 \sqrt{x^3}} \right)^{11}$
(A) there appears a term with the power x^2 (B) there does not appear a term with the power x^2
(C) there appears a term with the power x^{-3} (D) the ratio of the co-efficient of x^3 to that of x^{-3} is $1/3$

9. If it is known that the third term of the binomial expansion $(x + x^{\log_{10} x})^5$ is 10^6 then x is equal to-
(A) 10 (B) $10^{-5/2}$ (C) 100 (D) 5
10. Which of the following statement(s) is/are correct ?
(A) $1 + \frac{2}{2} + \frac{3}{2^2} + \frac{4}{2^3} + \dots + \infty = 4$
(B) Integral part of $(9 + 4\sqrt{5})^n$, $n \in \mathbb{N}$ is even.
(C) $({}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n)^2 = 1 + {}^{2n}C_1 + {}^{2n}C_2 + \dots + {}^{2n}C_{2n}$
(D) $\frac{1}{(3+2x)^2}$ can be expanded as infinite series in ascending powers of x only if $|x| < \frac{2}{3}$.
11. If $(9 + \sqrt{80})^n = I + f$ where I, n are integers and $0 < f < 1$, then -
(A) I is an odd integer (B) I is an even integer
(C) $(I + f)(1 - f) = 1$ (D) $1 - f = (9 - \sqrt{80})^n$
12. If for $n \in \mathbb{I}$, $n > 10$; $1 + (1+x) + (1+x)^2 + \dots + (1+x)^n = \sum_{k=0}^n a_k \cdot x^k$, $x \neq 0$ then
(A) $\sum_{k=0}^n a_k = 2^{n+1}$ (B) $a_{n-2} = \frac{n(n+1)}{2}$
(C) $a_p > a_{p-1}$ for $p < \frac{n}{2}$, $p \in \mathbb{N}$ (D) $(a_9)^2 - (a_8)^2 = {}^{n+2}C_{10} ({}^{n+1}C_{10} - {}^{n+1}C_9)$
13. Let $P(n) = \sum_{r=0}^n \frac{(-1)^r r^n}{r+1} C_r$. Now which of the following holds good ?
(A) $|P_{10}|$ is harmonic mean of $|P_9|$ & $|P_{11}|$ (B) $\sum_{r=5}^{10} P(r) P(r-1) = -\frac{6}{55}$
(C) $|P_{10}|$ is arithmetic mean of $|P_9|$ & $|P_{11}|$ (D) $\sum_{r=5}^{10} P(r) P(r-1) = \frac{6}{55}$
14. Let $(1+x)^m = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_mx^m$, where $C_r = {}^mC_r$ and $A = C_1C_3 + C_2C_4 + C_3C_5 + C_4C_6 + \dots + C_{m-2}C_m$, then -
(A) $A \geq {}^{2m}C_{m-2}$ (B) $A < {}^{2m}C_{m-2}$
(C) $A > C_0^2 + C_1^2 + C_2^2 + \dots + C_m^2$ (D) $A < C_0^2 + C_1^2 + C_2^2 + \dots + C_m^2$
15. Consider $E = (\sqrt[3]{x} + \sqrt[5]{y})^z = I + f$, $0 \leq f < 1$
(A) If $x = 5$, $y = 2$, $z = 100$, then number of irrational terms in expansion of E is 98
(B) If $x = 5$, $y = 2$, $z = 100$, then number of rational terms in expansion of E is 4
(C) If $x = 16$, $y = 1$ & $z = 6$, then $I = 197$
(D) If $x = 16$, $y = 1$ & $z = 6$, then $f = (\sqrt{2} - 1)^6$

EXERCISE # 6

- If $(7 + 4\sqrt{3})^n = p + \beta$ where n & p are positive integers and β is a proper fraction show that $(1 - \beta)(p + \beta) = 1$.
- Let $P = (2 + \sqrt{3})^5$ and $f = P - [P]$, where $[P]$ denotes the greatest integer function.
Find the value of $\left(\frac{f^2}{1-f}\right)$.
- For which positive values of x is the fourth term in the expansion of $(5 + 3x)^{10}$ is the greatest.
- Prove that $\sum_{K=0}^n {}^nC_K \sin Kx \cdot \cos(n-K)x = 2^{n-1} \sin nx$.
- Let $a = (4^{1/401} - 1)$ and let $b_n = {}^nC_1 + {}^nC_2 \cdot a + {}^nC_3 \cdot a^2 + \dots + {}^nC_n \cdot a^{n-1}$.
Find the value of $(b_{2006} - b_{2005})$
- Let a and b be the coefficient of x^3 in $(1 + x + 2x^2 + 3x^3)^4$ and $(1 + x + 2x^2 + 3x^3 + 4x^4)^4$ respectively. Find the value of $(a - b)$.
- Find the sum of the roots (real or complex) of the equation $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0$.
- Find the index n of the binomial $\left(\frac{x}{5} + \frac{2}{5}\right)^n$ if the 9th term of the expansion has numerically the greatest coefficient ($n \in \mathbb{N}$).
- Find the coefficient of x^{49} in the polynomial $\left(x - \frac{C_1}{C_0}\right)\left(x - 2^2 \cdot \frac{C_2}{C_1}\right)\left(x - 3^2 \cdot \frac{C_3}{C_2}\right) \dots \left(x - 50^2 \cdot \frac{C_{50}}{C_{49}}\right)$, where $C_r = {}^{50}C_r$.
- If $\binom{n}{r}$ denotes nC_r , then
 - Evaluate : $2^{15} \binom{30}{0} \binom{30}{15} - 2^{14} \binom{30}{1} \binom{29}{14} + 2^{13} \binom{30}{2} \binom{28}{13} \dots - \binom{30}{15} \binom{15}{0}$
 - Prove that : $\sum_{r=1}^n \binom{n-1}{n-r} \binom{n}{r} = \binom{2n-1}{n-1}$
 - Prove that : $\binom{n}{r} \binom{r}{k} = \binom{n}{k} \binom{n-k}{r-k}$

ANSWER KEY

EXERCISE # 1

- | | | | | | | |
|-------|------|-------|-------|-------|-------|-------|
| 1. C | 2. B | 3. A | 4. C | 5. B | 6. A | 7. C |
| 8. A | 9. B | 10. B | 11. B | 12. D | 13. B | 14. A |
| 15. B | | | | | | |

EXERCISE # 2

- | | | | | |
|---|--|----------------------|--------------------------|---|
| 1. (a) $r = 6$ | (b) $r = 5$ or 9 | 2.(i) $\frac{5}{12}$ | (ii) $T_6 = 7$ | 4. $\frac{(2^{mn} - 1)}{(2^n - 1)(2^{mn})}$ |
| 5. (i) $T_7 = \frac{7 \cdot 3^{13}}{2}$ | (ii) 455×3^{12} | 6. $\frac{17}{54}$ | 7. $n = 2$ or 3 or 4 | |
| 8. 816 | 10. (a) $-1260 \cdot a^2 b^3 c^4$; (b) -12600 | 11. (a) 990 (b) 3660 | | |
| 12. ${}^nC_r(3^{n-r} - 2^{n-r})$ | 14. (i) 3^n (ii) 1, (iii) a_n | | | |

EXERCISE # 3(JM)

- | | | | | | | |
|------------|-----------|-----------|-----------|------------|-------|-------|
| 1. 2 | 2. 1 | 3. 3 | 4. 2 | 5. 4 | 6. 3 | 7. 1 |
| 8. 2 | 9. 3 | 10. 4 | 11. 3 | 12.(Bonus) | 13. 1 | 14. 1 |
| 15. 4 | 16. 3 | 17. 4 | 18. 2 | 19. 4 | 20. 1 | 21. 3 |
| 22. 2 | 23. 3 | 24. 4 | 25. 4 | 26. 4 | 27. 2 | |
| 28. Bonus | 29. 1 | 30. 1 | 31. 1 | 32. 1 | 33. 4 | 34. 3 |
| 35. 3 | 36. 30.00 | 37. 2 | 38. 2 | 39. 4 | 40. 1 | |
| 41. 615.00 | 42. 3 | 43. 51.00 | 44. 2 | 45. 3 | 46. 4 | 47. 1 |
| 48. 3 | 49. 8.00 | 50. 3 | 51. 13.00 | 52. 120.00 | 53. 1 | 54. 3 |

EXERCISE # 4(JA)

- | | | | | | | |
|------|-------|------------|----------|-----------|------|------|
| 1. C | 2. A | 4. D | 5. A | 6. D | 7. 6 | 8. C |
| 9. 8 | 10. 5 | 11. 646.00 | 12. 6.20 | 13. A,B,D | | |

EXERCISE # 5

- | | | | | | | |
|--------|-------|--------|---------|---------|--------|--------|
| 1. B | 2. B | 3. B | 4. D | 5. A | 6. ACD | 7. AB |
| 8. BCD | 9. AB | 10. AC | 11. ACD | 12. BCD | 13. AD | 14. BD |
| 15. AC | | | | | | |

EXERCISE # 6

- | | | | | | |
|-----------|--------------------------------------|-------------|------|--------|-----------|
| 2. 722 | 3. $\frac{5}{8} < x < \frac{20}{21}$ | 5. 2^{10} | 6. 0 | 7. 500 | 8. $n=12$ |
| 9. -22100 | 10. (a) $\binom{30}{15}$ | | | | |

